

AI-Powered Mental Health Chatbot Using Large Language Models

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Abstract— This study describes the development and testing of an AI-powered chatbot for mental health that uses the most recent large language models (Llama, Qwen, and Gemma) to deliver sensitive, user-friendly, and context-sensitive emotional care. In order to bring the relevance of the response level and counseling-style regularity, this system integrates the upgraded versions of these models with a Retrieval-Augmented Generation pipeline. All of the models provided benefits, according to the exploratory investigation. ROUGE scores, which calculate the language overlap between model responses and reference counselor responses, and BERT scores, which gauge semantic and meaning similarity, are two quantitative and qualitative metrics offered by model evaluation. In conclusion, this study demonstrates how a scalable, low-barrier mental model may be developed by integrating many LLM families with RAG and safety-oriented generation.

Keywords— *Large Language Models, Mental Health Chatbot, Llama, Qwen, Gemma, Retrieval-Augmented Generation (RAG), Gen AI, Natural Language Processing.*

I. INTRODUCTION

Anxiety, sadness, and stress-related diseases are examples of mental health disorders that have sharply grown globally among people of all ages and backgrounds. People frequently find it difficult to acquire timely psychological care due to a dearth of mental health experts, lengthy waiting lists, financial hardships, and persistent social stigma [1]. In this regard, AI-based conversational systems have emerged as a quick, accessible, and different way to offer instant emotional support and lower obstacles to asking for assistance [2]. According to [3], traditional mental health chatbots are rule-based or template-driven, but their inflexible architecture guarantee minimal adaptability and the incapacity to maintain long-term contextual consistency.

Rather, LLM-powered systems can be refined on therapeutic conversational datasets and further enhanced by RAG processes to show counselor-like communication patterns with less hallucinations, as in [4].

A. Motivation

Despite the recent steep rise in mental health issues, access to professional care is severely hampered by a number of systemic issues, such as high fees, a lack of qualified therapists, lengthy waiting lists, and the ongoing social stigma associated with getting help [5]. There are few quick, reasonably priced options for emotional help for young adults and those living in underprivileged areas. As the gap between growing mental health requirements and healthcare institutions' ability to meet them grows, a scalable technological solution becomes an essential necessity.

B. Research Objectives

The deployment and assessment of an AI-powered mental health chatbot that makes use of refined versions of LLaMA, Qwen, and Gemma to provide sympathetic and context-sensitive emotional support is the main contribution of this article [6]. This study demonstrates how to improve the quality of therapeutic responses by fine-tuning those LLMs on counseling-style datasets. In order to improve grounding and lessen hallucinations, this is coupled with a RAG framework. In order to give a linguistic and semantic alignment with reference counselor responses, it takes into account the ROUGE and BERTScore metrics for model performance. Additionally, this study explores an architecture for real-time deployment scalable at practical levels for digital well-being applications and looks at some of the advantages and disadvantages of each model family in relation to mental health contexts [11].

C. Preliminary Observations

The majority of the answers are brief or comforting, and the emotional settings do varied greatly in length, complexity, and depth. In order to achieve consistent treatment responses, preprocessing and creating appropriate prompts will be crucial. According to preliminary research, commercial LLaMA and Qwen/Gemma models have strong linguistic fluency and reasoning abilities, but when used in their unprocessed state, they consistently lack empathy, emotional grounding, and safety systems. Pre-trained LLMs frequently produce generic or overconfident therapeutic assertions, according to early tests, highlighting the necessity of domain-specific fine-tuning and stringent response limits.

II. LITERATURE REVIEW

Recently, AI-driven conversational systems for mental health support have surfaced, and demand in this field is rising sharply on a global scale. Accessibility and scalability continue to be the most essential requirements for emotional support. Traditional chatbots have been used for practical purposes such as behavioral intervention, mood monitoring, and guided self-help; however, new developments in Large Language Models have opened the door to the possibility of sympathetic, context-sensitive speech [18]. The therapeutic potential of LLMs, their limitations, and the safety measures required when implementing such systems in delicate psychological contexts are all covered in recent studies.

A. Mental Health Chatbots and Digital Therapeutics

Early mental health chatbots, like ELIZA, and later systems like Woebot and Wysa, demonstrated the idea by automating a supportive dialogue using scripted or rule-based frameworks. In a controlled trial, Fitzpatrick et al. [3] showed that Woebot, which is based on CBT principles, dramatically decreased depressed symptoms. This implies that digital agents have the potential to offer significant psychological

advantages. Similarly, chatbot-guided self-help tools may be useful for enhancing emotional health and lowering perceived stress, according to Inkster et al. [11].

However, these conventional systems are mostly dependent on pre-established conversational flows, which restricts their ability to provide subtle therapeutic responses or adjust to complicated user emotions. Research toward more adaptable AI-driven conversational models [17] that can carry out fuller interactions is prompted by the fact that, despite providing structure and safety, their rigid design hinders deeper personalization.

B. Large Language Models in Mental Health Support

By enabling meaningful, contextually grounded, and human-like discourse, large language models like as LLaMA, Qwen, and Gemma [19] transformed natural language understanding. Large language models showed considerable promise in identifying risk, offering emotional support, and considering mental health, according to a comprehensive analysis by Guo et al. [4]. When compared to rule-based systems, the responses produced by LLMs are accompanied by improved context awareness and modified reasoning, which could make them better for conversational applications in mental health support. Recent studies show that refined big language models may validate emotions, adopt therapist-like reactions, and offer structured coping recommendations.

Additionally, researchers point out that unaltered LLMs frequently produce overconfident clinical assertions or hallucinations; hence, domain-specific fine-tuning is essential to guaranteeing safety alignment for responsible deployment.

C. RAG and Model Optimization

According to P. Lewis et al. [6], RAG is appropriate for sensitive fields like mental health assistance because it enhances knowledge consistency by integrating pertinent retrieved information into the creation process. Retrieval-guided decoding improves factuality and safety to help models avoid unsupported or potentially dangerous outputs, as demonstrated by Y. Liu et al. [7]. T. Zhang discovered that refined versions of LLaMA, Qwen, and Gemma variations can function at the level of domain-specialized bigger systems [9]. At the same time, instruction tweaking has improved empathy and contextual reasoning for open-source models. According to these findings, a straightforward method to generating grounded, sympathetic, and safety-aligned chatbot replies is to combine RAG with targeted fine-tuning.

D. Safety, Risk Detection, and Ethical Considerations

When using AI in mental health settings, safety is a top priority. The model's potential and limitations as a decision-support tool were demonstrated by Lee et al. [8], who demonstrated that GPT-4 could assess text-based complaints to identify individuals at risk of suicide with a performance comparable to physicians. In the absence of explicit risk-detection systems at multiple levels, LLMs may misidentify crisis circumstances, misinterpret emotional cues, or give dangerous advice through their outputs, according to numerous studies [16].

In order to prevent models from making clinical diagnoses or prescriptive treatment suggestions, ethical guidelines include layered safety frameworks that include filtering, crisis escalation channels, and limitations. When modifying open-source models like LLaMA, Qwen, and Gemma that lack the

built-in safeguards seen in proprietary systems, these safety concerns are especially important.

E. Research Gaps and Study Rationale

Despite growing research, several gaps persist in the development of AI mental health [17] chatbots. Many existing systems rely on proprietary LLMs, limiting transparency and preventing full safety customization. Few studies provide direct comparisons of open-source LLM families such as LLaMA, Qwen, and Gemma within mental health contexts. While safety concerns are widely acknowledged, there is limited empirical research integrating it. Existing evaluations often focus on linguistic quality rather than metrics aligned with therapeutic appropriateness, empathy, or semantic grounding. This study addresses these gaps by developing a multi-model mental health chatbot, fine-tuning open-source LLMs on counseling data, integrating RAG for improved grounding, and evaluating performance using ROUGE and BERTScore.

III. METHODOLOGY

A. Data Sources and Acquisition

The Amod/mental_health_counseling_conversations dataset, a publicly accessible text-based compilation of talks between seekers and counselors' responses, is the primary source of data used in this study. Because of its stable versioning, which guarantees repeatability across experiments, the dataset was acquired via the HuggingFace repository. It contains a variety of conversational situations that arise in daily life, ranging from stressful emotions to more difficult psychological situations, and it serves as a suitable foundation for teaching models meant to react appropriately and empathetically. As a result, these sources offer a varied and moral corpus for the creation and improvement of the AI chatbot for mental health.

B. Data Preprocessing

Cleaning and standardizing the dataset to produce high-quality input was part of data preparation. Correcting irregular spacing, eliminating non-ASCII letters, and getting rid of unnecessary punctuation or formatting artifacts were all part of normalizing text entries. Every conversation was arranged into structured contexts with pairs of responses. After that, helpful characteristics like token counts and message length patterns were identified through feature engineering. In order to determine the sequence lengths for the models, the context and response length distributions were next investigated. In order to provide a clean, consistent corpus for pre-training, fine-tuning, and data evaluation, the dataset was finally divided 70% for training and 30% for testing following pre-processing.

C. Model Selection and Computational Analysis

To evaluate differences that might affect performance in mental health dialogue situations, three transformer-based language models LLaMA-3.2-1B, Gemma-3-1B-IT, and Qwen2-0.5B-Instruct have been chosen based on their different architectural size and computational complexity. The models have different numbers of parameters, hidden dimensionalities, and FLOPs per forward pass, although having comparable high-level architectures.

Coherent long-form reasoning is possible using the **LLaMA-3.2-1B** model, which contains roughly 1 billion parameters and a moderately large hidden size. Theoretically,

the model uses the basic transformer compute estimate to train and infer: The per-layer complexity of its transformer architecture is approximately

$$\text{FLOPs}_{\text{LLaMA}} \approx 2L(n^2d + nd^2) \quad (1)$$

where L is the number of layers, n the sequence length, and d the hidden dimension. LLaMA performs well on contextual reasoning because of its higher d , although it costs more for longer sequence lengths.

Additionally in the 1B parameter range, the **google/gemma-3-1b-it** model is tuned for instruction-following with effective low-precision inference. Gemma uses simplified attention techniques and less memory cost, resulting in lower practical FLOPs during inference, even though it has a similar number of parameters to LLaMA. Despite having a theoretical complexity that is comparable to LLaMA, this makes Gemma computationally appealing for deployments where reliable latency and resource efficiency are crucial.

where d_g (Gemma's hidden size) is slightly smaller and more optimized for low-precision inference (BF16/FP8). As a result, Gemma achieves lower effective computation per token than LLaMA despite similar parameter count.

The **Qwen/Qwen2-0.5B-Instruct** model, with approximately 500 million parameters, offers the lowest computational footprint among the selected architectures. Its inference cost scales roughly as

$$\text{FLOPs}_{\text{Qwen}} \propto 2L(n^2d) \quad (2)$$

where both L and $d_q < d_g < d_l$, yielding the smallest FLOPs among the selected models. Qwen therefore offers the fastest inference and lowest memory footprint, making it well suited for real-time deployment on limited hardware.

Together, these models represent three computational tiers full small-scale (1B), optimized small-scale (1B-it), and ultra-lightweight (0.5B) allowing a structured comparison of empathy quality, grounding reliability, and safety alignment across architectures.

D. Model Architecture Analysis

The suggested chatbot's overall architecture is based on a modular transformer-based pipeline, which is intended to guarantee that responses are sensible, sympathetic, and secure. One of three readily interchangeable LLMs; LLaMA-3.2-1B, Gemma-3-1B-IT, and Qwen2-0.5B-Instruct [15] will serve as the foundation for each system. Each system's fundamental generating component is these models, where it is converted into numerical token sequences of length n , forming the input matrix $X \in \mathbb{R}^{n \times d}$, where d is the model-specific embedding size. The standard transformer architecture, which consists of L stacked layers of multi-head self-attention and position-wise feed-forward modules, is followed by each chosen LLM. The following formula is used by the self-attention mechanism to calculate contextualized representations:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

with computational complexity driven by the quadratic term $O(n^2d)$. This is followed by a feed-forward block with complexity $O(nd^2)$, jointly forming the FLOPs estimates previously described. Although the models share the same architectural backbone, differences in d , L , and attention head

count result in distinct computational profiles and inference behaviors.

E. Model Pre-Training

A domain-adaptive pre-training step was employed to acclimate the chosen LLMs to the linguistic style and emotional patterns found in mental health therapy. The typical causal language modeling objective, which minimizes the next-token prediction loss, was then used to train each model on the preprocessed dataset.

$$\mathcal{L}_{\text{LM}} = -\sum_{t=1}^n \log P(x_t | x_{<t}) \quad (4)$$

To enable efficient training and inference on limited hardware, all models were loaded with 4-bit quantization using BitsAndBytes, with computation performed in bfloat16. Key generation parameters included temperature=0.7, top-p=0.9, max_new_tokens=150, and use_cache=False was applied to prevent CUDA in-place update conflicts. This configuration ensured reduced memory usage, faster optimization cycles, and stable generation quality. This phase enabled LLaMA-1B, Gemma-1B-IT, and Qwen-0.5B-Instruct to internalize supportive dialogue structures and empathetic phrasing prior to supervised fine-tuning, providing a stronger foundation for generating emotionally aligned mental health responses.

F. Model Fine-Tuning

Fine-tuning was carried out using a parameter-efficient approach that adapts each model to counseling-style dialogue without modifying the full set of pre-trained weights. This process relies on low-rank adaptation, where only a small update matrix is learned according to

$$W = W_0 + BA \quad (5)$$

with W_0 representing the frozen model weights and BA the trainable low-rank update, where $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$ are low-rank matrices with rank $r \ll d$. The combined optimization strives to minimize the supervised fine-tuning loss:

$$\mathcal{L}_{\text{SFT}} = -\sum_{t=1}^n \log P(x_t^{\text{response}} | x_{<t}^{\text{context}}; W_0, BA) \quad (6)$$

In this setup, LLaMA-3.2-1B benefits from improved long-form coherence, Gemma-3-1B-IT refines its instruction-tuned behavior for supportive communication, and Qwen2-0.5B strengthens its contextual and emotional alignment despite having fewer parameters.

1) QLoRA

QLoRA quantizes the base model to 4-bit quantized [18] training setup to efficiently adapt the LLMs to counseling-style responses. The models were loaded in NF4 4-bit precision with bfloat16 computation, reducing GPU memory use by up to 75%, while maintaining numerical stability. This enables fine-tuning of billion-parameter models on modest hardware.

2) LoRA

LoRA injects low-rank matrices into attention and feed-forward layers, allowing the model to learn task-specific adjustments while preserving the pre-trained general knowledge of the backbone. The LoRA configuration used a rank of $r=16$, scaling factor $\alpha=32$, and dropout 0.1, enabling the model to learn task-specific behavior without modifying the full parameter set.

G. Retrieval-Augmented Generation (RAG)

RAG was employed to enhance grounding and reduce hallucination [16] by retrieving counseling examples that are semantically similar to the user's query. Each query q is embedded using a sentence-transformer to obtain $e_q = f(q)$, and the system retrieves the top- k nearest contexts by minimizing cosine distance,

$$d(e_q, e_i) = 1 - \frac{e_q \cdot e_i}{\|e_q\| \|e_i\|} \quad (7)$$

The retrieved contexts and their corresponding counselor responses are then integrated into a structured prompt designed to guide generation without enabling copying. Formally, the augmented model input is

$$X_{\text{RAG}} = [q; R] \quad (8)$$

where R indicates the examples that were retrieved. In actuality, a Chroma vector database was used to embed and index every training scenario. This made the similarity search during inference more effective. The system retrieved the three most comparable answers for a particular test query and added them to the prompt template. The LLaMA, Gemma, or Qwen model produced an answer based on the inquiry and stylistic direction. Retrieval-driven prompting like this improves contextual relevance, softens the responses' tone, and moves the model's output closer to counseling communication patterns.

H. Workflow Diagram

The suggested mental health chatbot system's whole pipeline, from data preparation to model inference, is summed up in the workflow diagram.

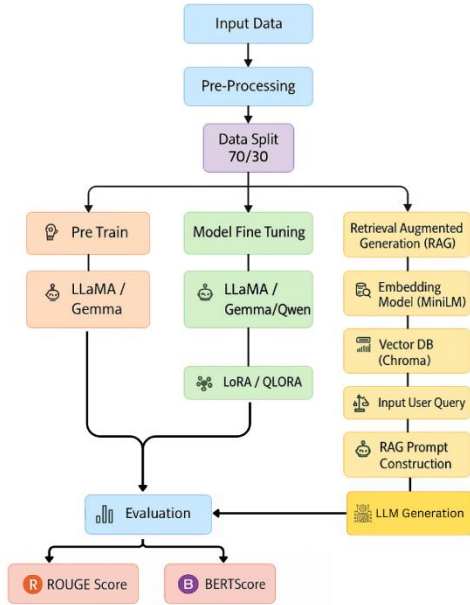


Fig. 1. Workflow Diagram

It explains, step-by-step, how the different training, retrieval, and generation components ultimately result in a grounded, sympathetic, and secure chatbot answer.

IV. RESULTS AND ANALYSIS

An examination of the model's performance in relation to pre-training, fine-tuning, and RAG is shown in this section [16]. Both quantitative and qualitative results are used to assess the responses' quality, grounding, and semantic similarity. There are comparative findings that demonstrate the behavior of each model for various configurations. These results offer a clear understanding of the extent to which retrieval augmentation and parameters have effectively tuned the models to produce precise, sympathetic, and contextually relevant replies regarding mental health.

A. Performance of LLaMA-3.2-1B

1) Pre-Training Analysis

Strong semantic similarity and moderate lexical alignment with the counseling responses were attained during the pre-training phase. With ROUGE-1, ROUGE-2, and ROUGE-L scores of 0.2657, 0.0384, and 0.1439, respectively, the model demonstrated some overlap with reference phrase. Despite its limited surface level similarity, a significantly higher BERTScore of 0.8314 indicates that the model successfully caught the underlying semantic and emotional intent of the replies. These findings demonstrate that pre-training effectively establishes a semantic basis, but more refinement is needed to produce outputs that are more accurate and artistically coherent.

2) Fine-Tuning Analysis

Training and validation losses decreased over the course of the epochs as a result of fine-tuning: Epoch 1 was 2.2741/2.2344, Epoch 2 was 1.9396/2.0320, and Epoch 3 was 1.6798/1.9704. As a result, the model consistently fitted the dataset's counseling-style patterns. As a result, the assessment metrics scores improved: ROUGE rose to 0.2946 for ROUGE-1, 0.0465 for ROUGE-2, and 0.1511 for ROUGE-L, suggesting a stronger lexical and structural match with target replies. The BERTScore F1 rising to 0.8392 confirms greater semantic similarity and more accurate meaning preservation.

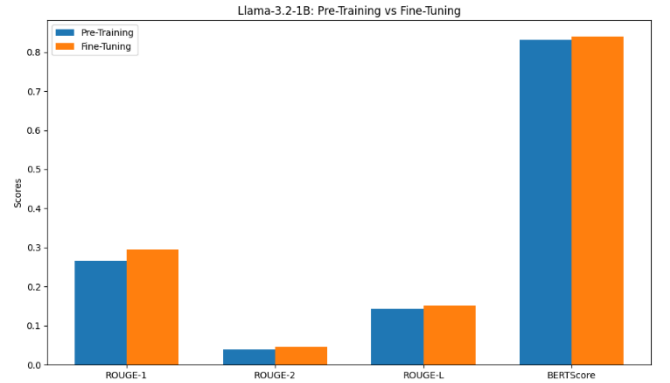


Fig. 2. Llama 3.2-1b Pre Training vs Fine-Tuning

3) RAG Analysis

The fact that the answers were more firmly based on pertinent counseling examples serves as proof of this. Lexical alignment remained stable, as evidenced by ROUGE scores of 0.2871, 0.0397, and 0.1360, which are comparable to the refined model. The model draws from the retrieved samples rather than precisely matching the reference responses, as evidenced by the BERTScore of 0.7561, which indicates a shift in semantic structure. RAG successfully improved contextual correctness, coherence, and decreased hallucinated

content despite the lower semantic similarity score, hence improving the model's capacity to provide answers.

B. Performance of Gemma-3-1b-it

1) Pre-Training Analysis

During the pre-training phase, the Gemma-3-1b model has a very strong alignment with the counseling dataset. For lexical and structural similarity to the target replies, ROUGE scores of 0.3039 for ROUGE-1, 0.0503 for ROUGE-2, and 0.1471 for ROUGE-L are reasonable and marginally higher than in the LLaMA pre-trained baseline. This is supported by a BERTScore of 0.8356, which demonstrates strong semantic alignment and shows that Gemma successfully grasped both meaning and emotional purpose provided in counselor-style language foundation. However, additional refinement is still required for more accurate and stylistically aligned outputs.

2) Fine-Tuning Analysis

The progressive decline in training and validation losses throughout epochs (Epoch 1: 2.2547/2.1904, Epoch 2: 1.9458/2.0079, and Epoch 3: 1.7165/1.9456) indicates incremental but mid-level continuous gains in Gemma-1B fine-tuning. Put otherwise, it indicates that the model primarily learned from the counseling style dataset's pattern. These enhancements are demonstrated by the ensuing performance measures, which exhibit mid-range improvements in lexical accuracy and structural alignment with ROUGE scores of 0.3060 for ROUGE-1, 0.0493 for ROUGE-2, and 0.1429 for ROUGE-L. Strong semantic similarities are shown by the BERTScore F1 of 0.8326, which offers more sophisticated and context-sensitive outputs while staying near pretraining values.

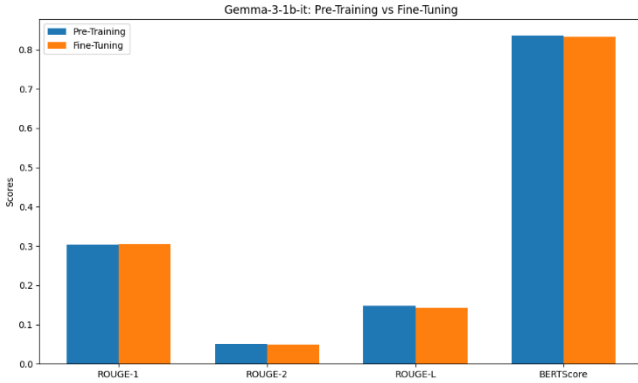


Fig. 3. Gemma 3-1b Pre Training vs Fine-Tuning

3) RAG Analysis

The RAG-enhanced Gemma model's lexical overlap did not change, but there was a clear semantic shift. ROUGE scores of 0.2907, 0.0412, and 0.1380 show similar phrase-level similarity to reference responses and stay near the refined baseline. However, as the model relies more and more on retrieved instances to build responses, semantic closeness decreases with a BERTScore of 0.7517. It shows that RAG enhances contextual grounding and relevance, providing more targeted and encouraging responses even when similarity measurements fall short of capturing these improvements.

C. Performance of Qwen2-0.5B-Instruct

1) Pre-Training Analysis

During pre-training, Qwen2-0.5B-Instruct's alignment with the counseling dataset was moderate. Although the

ROUGE scores of 0.2946 (ROUGE-1), 0.0369 (ROUGE-2), and 0.1369 (ROUGE-L) indicate that some surface-level structure was recorded, the responses' proximity to reference phrasing was still rather limited. Qwen learned the broad meaning and tone of counseling discussion but failed to fully catch stylistic details, as evidenced by the BERTScore of 0.8361, which indicates semantic similarity is strong but not significantly higher than the larger models.

2) Fine-Tuning Analysis

According to the training and validation losses over epochs, it has also led to consistent optimization improvements for Qwen2-0.5B-Instruct. In spite of this, the model's evaluation ratings revealed that, when compared to LLaMA and Gemma, the performance increases from it remained comparatively low. ROUGE scores of 0.2750 (ROUGE-1), 0.0375 (ROUGE-2), and 0.1349 (ROUGE-L) fall short of the refined outcomes from the larger models and are marginally below their pre-training values. According to the training and validation losses over epochs, it has also led to consistent optimization improvements for Qwen2-0.5B-Instruct. In spite of this, the model's evaluation ratings revealed that, when compared to LLaMA and Gemma, the performance increases from it remained comparatively low. ROUGE scores of 0.2750 (ROUGE-1), 0.0375 (ROUGE-2), and 0.1349 (ROUGE-L) fall short of the refined outcomes from the larger models and are marginally below their pre-training values.

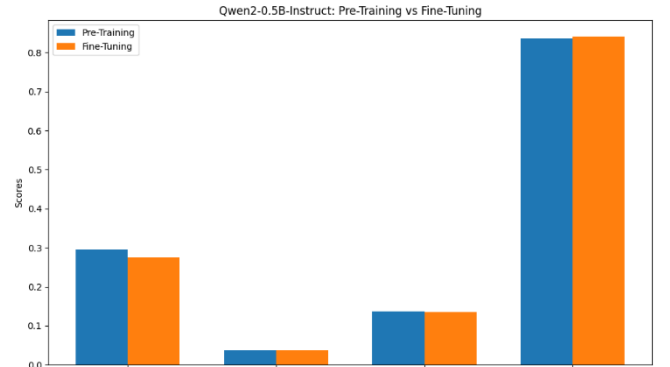


Fig. 4. Qwen 2-0.5b Pre Training vs Fine-Tuning

3) RAG Analysis

The RAG configuration produced inconsistent results with Qwen2-0.5B-Instruct. For the remainder, its quantitative performance remained relatively poor, but it got somewhat more anchored in context. ROUGE scores (0.2807, 0.0360, and 0.1322) were just little different from the refined baseline, indicating that RAG also performed poorly in terms of lexical correctness. However, semantic similarity significantly declined, as shown by a lower BERTScore of 0.7606, indicating that Qwen had difficulty integrating the data it acquired.

D. Comparative Performance Analysis Across Models

Based on their relative parameter sizes and architectural capability, the models show performance disparities that are rather well aligned across pre-training, fine-tuning, and RAG. Qwen2-0.5B-Instruct showed the lowest score, which is especially noticeable under RAG due to its reduced capacity limits retrieval integration. LLaMA-3.2-1B performs the best, while Gemma-1B-IT stays at a more stable midtier model. The comparison chart below effectively illustrates these patterns.

TABLE I. MODEL PERFORMANCE COMPARISON

Model	Setting	R-1	R-2	R-L	BERT	Overall Performance
LLaMA-3.2-1B	Fine-Tune	0.2946	0.0465	0.1511	0.8392	High
LLaMA-3.2-1B	RAG	0.2871	0.0397	0.1360	0.7561	High (RAG lowers semantic similarity)
Gemma-3-1B-IT	Fine-Tune	0.3060	0.0493	0.1429	0.8326	Mid-High
Gemma-3-1B-IT	RAG	0.2907	0.0412	0.1380	0.7517	Mid-High (RAG stable)
Qwen2-0.5B-instruct	Fine-Tune	0.2750	0.0375	0.1349	0.8405	Mid-Low (weaker structure)
Qwen2-0.5B-instruct	RAG	0.2807	0.0360	0.1322	0.7606	Low (difficulty using retrieved context)

E. Other Insights:

In addition to the quantitative measurements, a number of qualitative trends were found in all of the models. Gemma's responses were typically brief but steady with a narrow emotional range, whereas LLaMA consistently produced the most coherent and stable responses, particularly for ambiguous prompts. Although effective, Qwen was inconsistent and tended to respond in a way that was too cautious or generic when the user's meaning was unclear. Each model exhibited varying degrees of emotional cue sensitivity; LLaMA responded more adaptively, whereas Qwen had the strongest propensity to return to minimally safe responses.

Another finding was that there was a trade-off between the depth of linguistics and the level of computation: LLaMA required more computation but was able to provide richer and more context-aware responses, whereas Qwen provided the fastest inference but had the least expressive responses. These variations highlight the ways in which architectural capacity affects mental-health dialogue systems' stability, emotional response, and conversational dependability.

V. DISCUSSION AND IMPLICATIONS

By going over model behaviors, user experience considerations, ethics, and real-world deployment, this part expands on the experimental findings. The findings presented here go beyond quantitative measurements to show how effectively models function in actual conversational contexts of AI systems for mental health.

A. Conversational Reliability and Stability

One important finding is that, in situations where the cues were unclear, LLaMA was the most dependable in obtaining emotionally consistent responses, whereas Gemma stayed stable but frequently resorted to simplifications to express her ideas in much fewer lines. With its smaller design, Qwen exhibited greater variability and sometimes resorted to overly cautious or minimalist replies. These discrepancies imply that conversational stability is heavily influenced by model capacity and should be considered when choosing systems for delicate mental health situations.

B. User Perception and Empathy Dynamics

The models' behavior have implications for how users may perceive empathy and support. The longer, more

detailed answers provided by LLaMA may foster a sense of awareness that promotes user participation in and candor about their emotional experience. Users who prefer straightforward, no-nonsense guidance but may feel emotionally abandoned have found Gemma's shorter, more practical responses appealing. Although Qwen's conservative output runs the danger of coming out as detached or impersonal, it may enhance the protection against harm associated with responses. Understanding these stylistic variations is crucial since user comfort, trust, and satisfaction are important aspects of mental health support settings.

C. Robustness to Unstructured or Emotionally Complex Inputs Implications

Models' performance on dataset-aligned inquiries was predictable, but they differed significantly in their capacity to respond to unexpected, emotionally charged, or incomplete prompts. LLaMA demonstrated the most flexibility, frequently requesting clarifications or offering subtle assistance. In these situations, GEMMA tended to revert to general assurance, suggesting a relative stiffness in emotional expressiveness. Under these circumstances, QWEN exhibited the most instability and occasionally failed to sufficiently interact with the underlying emotional content. This highlights a crucial issue: systems designed for mental health applications must be stress-tested across a broad range of varied and unpredictable prompts because real users might not behave in accordance with structured patterns.

D. Ethical and Safety Considerations

According to the study, consumers may already perceive responses from technically sound models as emotionally authoritative. This is especially sensitive in situations involving mental health because there is a significant danger of misinterpretation or false confidence. One tactic for minimizing danger is reflected in Qwen's propensity for excessively cautious or repeating safety advisories, but it can also lower user trust or participation. Although encouraging, LLaMA's more expressive responses need to be closely watched to prevent unintentional advise. These findings highlight the necessity of multi-layered safety frameworks, transparent information about technology limitations, and stringent guidelines for when human professionals should be consulted.

E. Deployment Implications for Real-World Use

Naturally, the models differ in terms of practicality, not just in terms of performance but also in terms of computational footprint and operational viability. Gemma strikes a good balance between efficiency and quality, making it ideal for lightweight or embedded applications, while LLaMA offers the best interaction quality but requires more processing power, making it more appropriate for a hosted clinical tool than for an edge device. Qwen's low resource requirements make it especially suitable for deployment under extreme hardware constraints, but its conversational limits indicate that it would be more appropriate as an auxiliary assistance than a primary mental health responder. These deployment trade-offs further highlight the need for model selection to align with both therapeutic goal and technical competence.

VI. CONCLUSION AND FUTURE WORK

This final section summarizes the key findings, highlights the contributions of the study, and outlines potential directions for future research in AI-mental health chatbot.

A. Summary of Findings

In terms of producing coherent, sympathetic, and context-aware mental-health responses, this study shows that LLaMA-1B performs the best overall; Gemma-1B offers stable mid-level performance with efficient output; Qwen-0.5B, despite being lightweight, struggles with stylistic depth and consistency; fine-tuning improves semantic alignment across all models; RAG reinforces contextual grounding but does so with noticeable unevenness, especially for smaller architectures; and in conclusion, model capacity has a significant effects on conversational stability and emotional sensitivity.

B. Contributions

In addition to offering a comprehensive pipeline for pre-training, fine-tuning, and retrieval-augmented generation tailored to counseling dialogue, the work advances the study of mental-health AI by directly comparing the performance of three lightweight, open-source LLMs in terms of their advantages and disadvantages in supportive communication. Safety patterns provide a deeper understanding of how various designs manage delicate interactions, and the work adds to our understanding of model behavior and emotional reactivity. Finally, it offers useful advice on creating trustworthy and responsible AI mental health support systems.

C. Limitations

The study uses standardized counseling data, which might not fully capture the variety of real-world mental health discussions. Tone quality and empathy are not adequately captured by automatic criteria such as ROUGE and BERTScore. Only English was used to test the models. Rather than only dataset quality, architectural constraints may also limit the generalizability of smaller models.

D. Future Work

Future research will concentrate on integrating the chatbot with all-encompassing agentic AI skills, such as the capacity to organize, analyze, and oversee multistep support plans as opposed to producing discrete responses. This includes creating agents with proactive support paths, tailored assistance, and long-term conversational memory. Additional areas of focus include enhancing the system's crisis-level interaction management capabilities by including risk-intent classifiers, specialized training data, and reinforcement learning to better detect suicidal ideation and acute emotional distress. For improved high-risk grounding, high-level retrieval and reasoning techniques such safety-first decision layers, adaptive filtering, and recursive RAG will be investigated. In order to guarantee safe, moral, and dependable implementation in actual mental health settings, future research may also focus on improving bigger or clinically aligned models, integrating them with therapist feedback loops, and creating hybrid human-AI escalation frameworks.

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